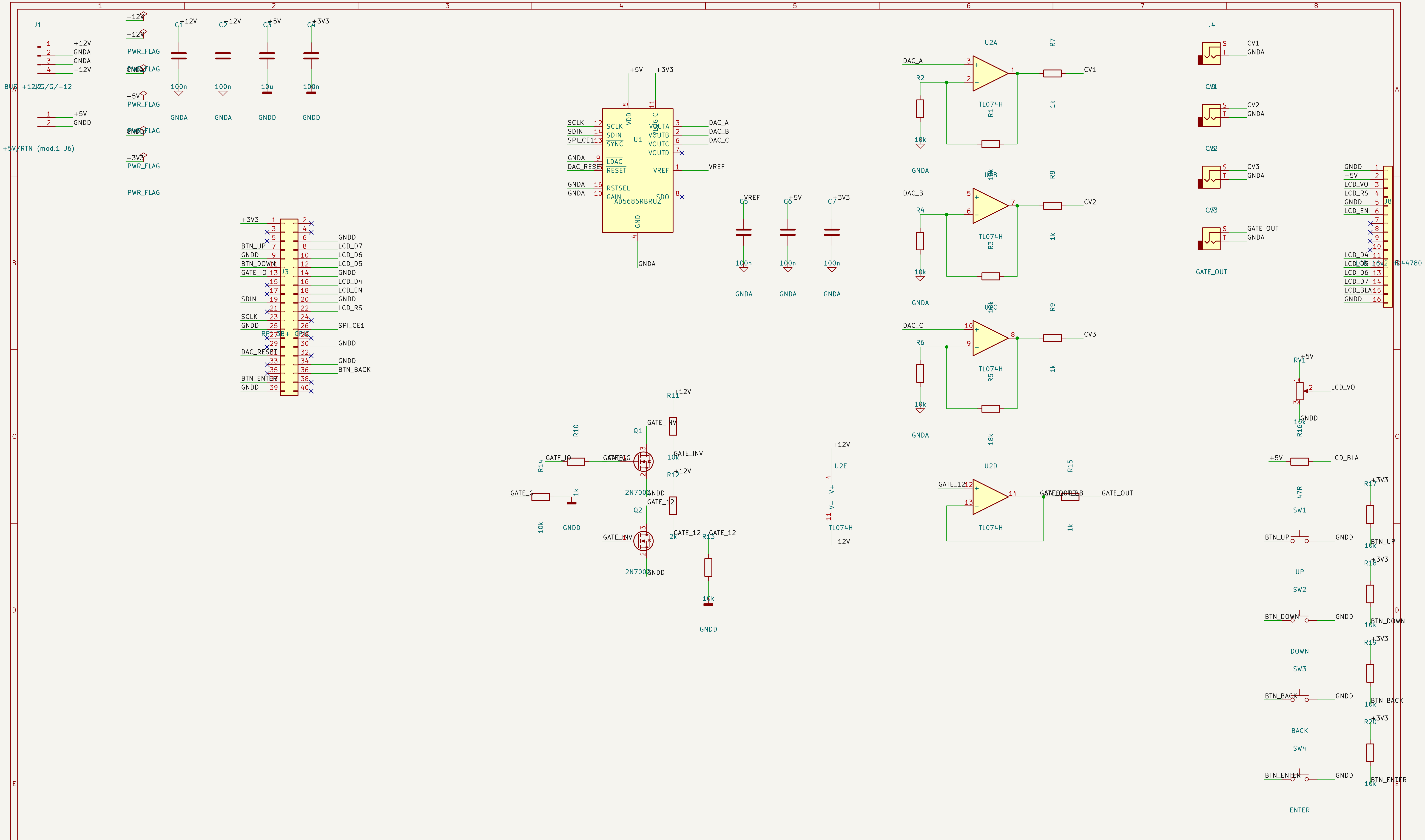
J1
+12V
GNDA
GNDA
-12V
BUS +12/G/G/-12
+5V
GNDD
+5V/RTN (mod.1 J6)
PWR_FLAG
C1 C2 C3 C4
100n 100n 10u 100n
GNDA GNDA GNDD GNDD
U1
AD5686RBRUZ
SCLK SDIN SYNC LDAC RESET RSTSEL GAIN
VDD VLOGIC GND
VOUTA VOUTB VOUTC VOUTD VREF SDO
SCLK SDIN SPI_CE1 GNDA DAC_RESET GNDA GNDA
DAC_A DAC_B DAC_C VREF
+5V +3V3
C5 C6 C7
100n 100n 100n
J3
RPi 3B+ GPIO
+3V3
BTN_UP GNDD BTN_DOWN GATE_IO SDIN SCLK GNDD DAC_RESET BTN_ENTER GNDD
GNDD LCD_D7 LCD_D6 LCD_D5 GNDD LCD_D4 LCD_EN GNDD LCD_RS SPI_CE1 GNDD GNDD BTN_BACK
U2A U2B U2C U2D U2E
TL074H
R1 R2 R3 R4 R5 R6 R7 R8 R9
10k 18k 1k
CV1 CV2 CV3
Q1 Q2
2N7002
R10 R11 R12 R13 R14 R15
GATE_IO GATE_G GATE_INV GATE_12 GATE_OUT
+12V -12V
J4 J5 J6 J7
CV1 CV2 CV3 GATE_OUT
J8
HD44780
GNDD +5V LCD_VO LCD_RS LCD_EN LCD_D4 LCD_D5 LCD_D6 LCD_D7 LCD_BLA
RV1
10k
R16
47R
SW1 SW2 SW3 SW4
UP DOWN BACK ENTER
BTN_UP BTN_DOWN BTN_BACK BTN_ENTER
R17 R18 R19 R20
10k
+3V3

GATE con dos MOSFET para que el reposo sea BAJO al arrancar
CV 0-7 V (la tesis usa 3-10). Presupuesto en ../tuning_budget.py
VLOGIC del DAC a +3V3, no a +5V: si no, VIH=3.5V y el Pi no llega
GNDA (bus, J1) y GNDD (retorno dedicado, J2) NO se unen en esta placa

SAM17 — Sintetizador analogico modular

Sheet: /
File: sam17-midicv.kicad_sch

Title: Modulo 02 — MIDI/CV: DAC 16 bits, 3x CV 1V/oct, GATE

Size: A3 Date: 2026-08-15 **Rev: A**
KiCad E.D.A. 10.0.3 Id: 1/1